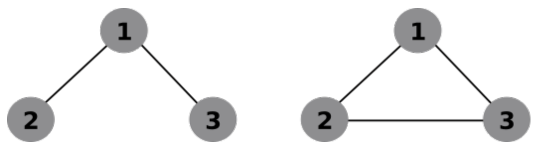
Machine Learning Exercise 12

Exercise 1

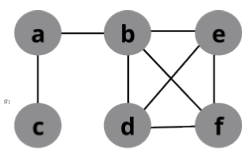
In the lecture we have defined two string kernels K_D in the form of a convolution kernel (String Kernels 1 and 2). We found that Kernel 2 is the same as the n-spectrum kernel we have previously encountered (Lecture 3, slide 23). What is the equivalent definition of convolution Kernel 1 in terms of a feature vector (similar to the features $\phi_{\boldsymbol{u}}, \phi_{\boldsymbol{u}}^{+}$ we have defined in lecture 3)?

Exercise 2

For the two graphlets g_1, g_2 of size 3:



and the graph G:

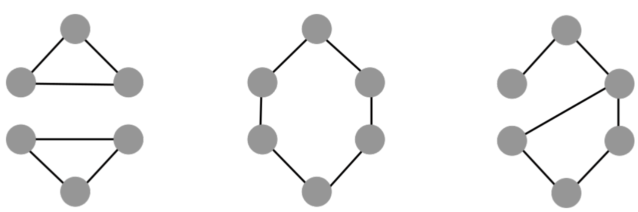


G

compute the feature values $\phi_{g_1}^{(1)}(G), \phi_{g_2}^{(1)}(G), \phi_{g_1}^{(2)}(G), \phi_{g_2}^{(2)}(G)$

Exercise 3

Recall the three graphs from slide 22:



A. Perform the first three rounds of WL code refinement:

- Start with each node having assigned the same code value
- Update (three times) according to the rule on slide 21. You can freely choose how you want to represent code values c^k (the code format on the slide is just an example), but you have to be sure that codes are unique in that the *hash* function returns different code values for different inputs.

B. Construct another graph for which the code values of all nodes will be the same in each iteration as for the nodes in the first two graphs above.